AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 16 · STUDENT LESSON

Overfitting Intuition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Audit the errors. Compare on validation. Limit the conclusion.

Separate numerical prediction error from accuracy and inspect model flexibility with complete validation records.

Keep one habit: Metric and units → supplied summaries → per-record MAE audit → controlled development and independent tie-break.

MAIN LESSON · SECTION 1

Does perfect training fit make a model the best choice?

> Use separate development evidence and a declared comparison rule.

Your learning goals

I can compare **training** and validation **evidence** for model **capacity**.

I can explain **overfitting** without assuming every gap proves it.

I can use a declared selection rule instead of perfect **training** fit.

Recall the limit of zero training error

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- Can a zero **training** error guarantee good validation performance?

Your end goal

By the end, I can use development performance rather than **training** fit alone to choose model **complexity**.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Choose Informative Questions. Give an example and a mistake that the idea helps you avoid.

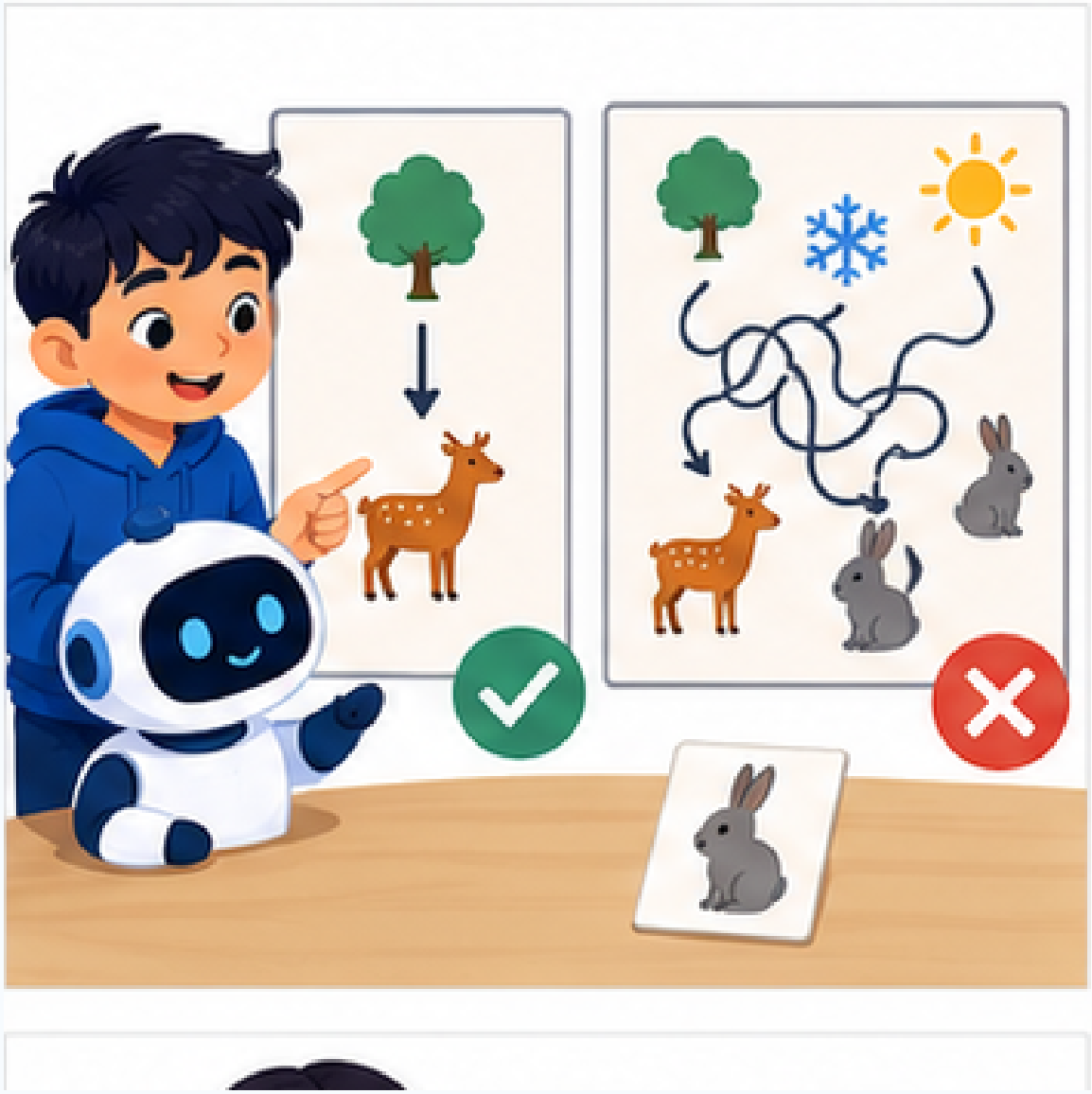
Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Human Classifier Simulation. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 16: **Overfitting** Intuition

A rule that memorizes examples may fail on new cases.
</details>



THINKER MISSION

Contrast a simple rule with a seven-**exception** rule.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

This week's end goal

MAIN LESSON · SECTION 2

Compare model flexibility using the same task and metric

> A training score and a validation score answer different questions.

Introduce capacity and separate evidence

A more flexible model can represent complicated relationships and accidental details. **Capacity** can help when there is enough relevant **evidence**, but perfect **training** performance is not the goal.

Compare **training** and validation using the same metric while checking leakage and representativeness.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **overfitting** and model **capacity**.

Name the numeric target, units and error direction

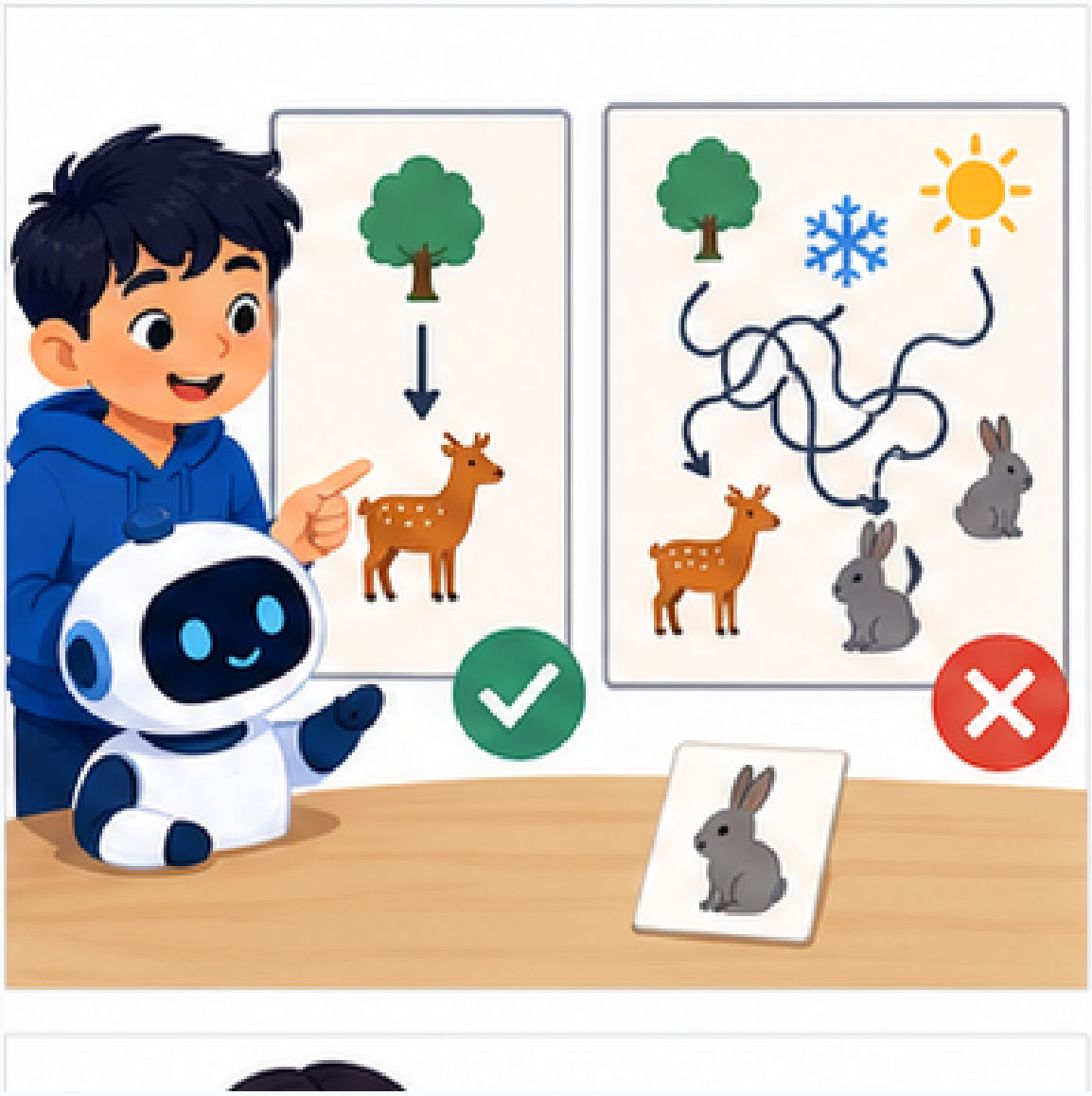
This week predicts a number, book length in pages, rather than a category. Use mean absolute error (MAE): take the absolute size of each prediction minus its reference, add those sizes and divide by the number of evaluated books. Lower is better; MAE0 means no error on those particular records. This is not the same metric as percent-correct accuracy, where higher is better.

▼ **Earlier wording, alternatives and picture transcripts**

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use **overfitting** intuition to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 3

Read the declared lowest-validation-error comparison

> These are supplied fictional model results, not a model-fitting run.

Define MAE and state the comparison rule

MAE means mean absolute error: find the size of each prediction mistake without its plus or minus sign, then average those sizes. Lower MAE means smaller average mistakes in the stated units. On the same regression split, a constant has **training** MAE 5 and validation 6, a line has **training** 2 and validation 3, and a highly flexible curve has **training** 0 and validation 8. Under smallest validation MAE, choose the line.

The curve has the lowest **training** loss but the worst validation result.

Keep training and validation in separate columns

Candidate	Training MAE (pages)	Validation MAE (pages)
Constant	5	6
Line	2	3
Flexible curve	0	8

Explain exactly which evidence supports the choice

- Before continuing, underline the decision rule and circle the **evidence** that supports the conclusion. Explain the example to a partner without replacing the counts or conditions with a stronger claim.

  Keep the distinction between an observed result and a proposed next step.

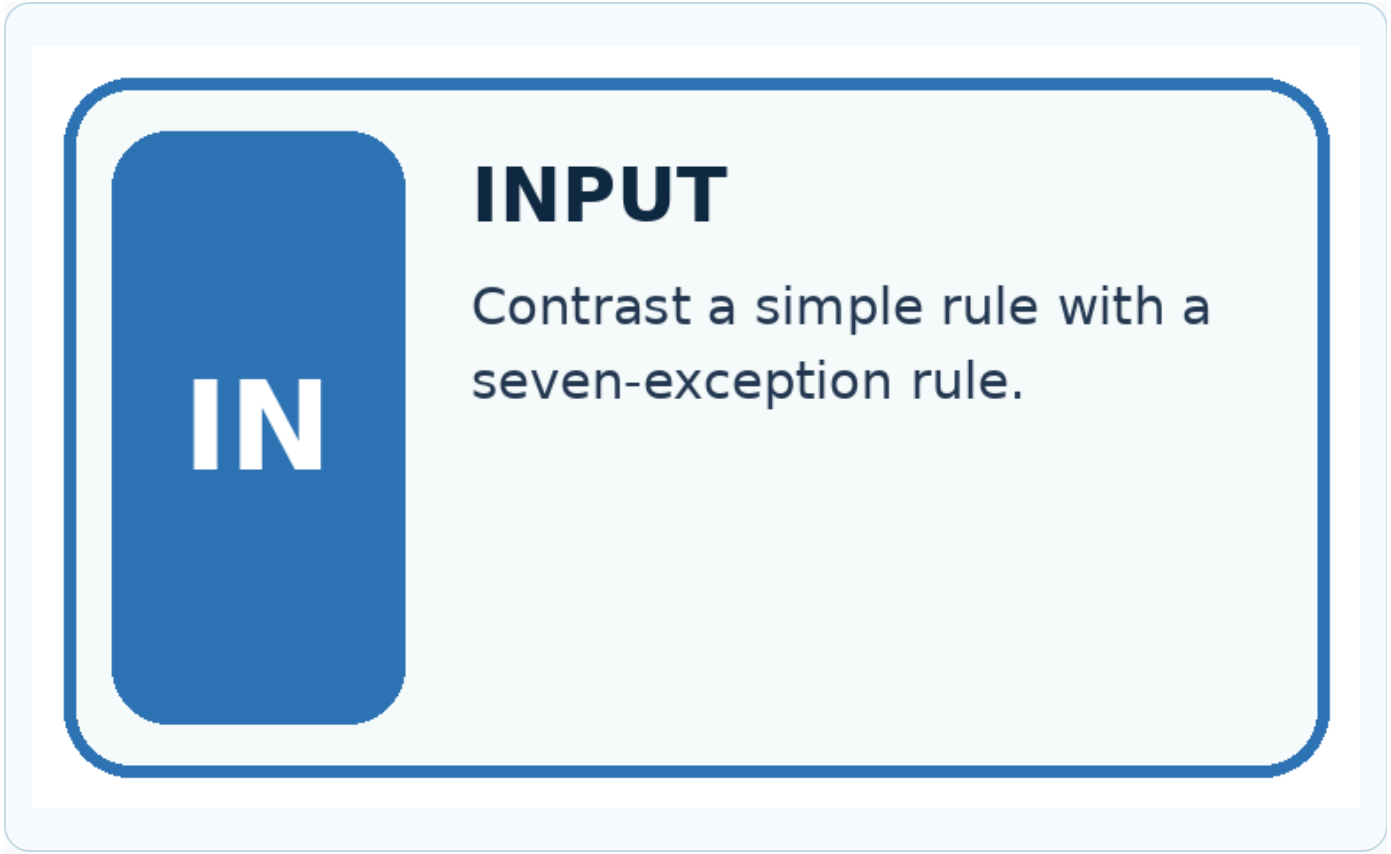
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Earlier wording, alternatives and picture transcripts

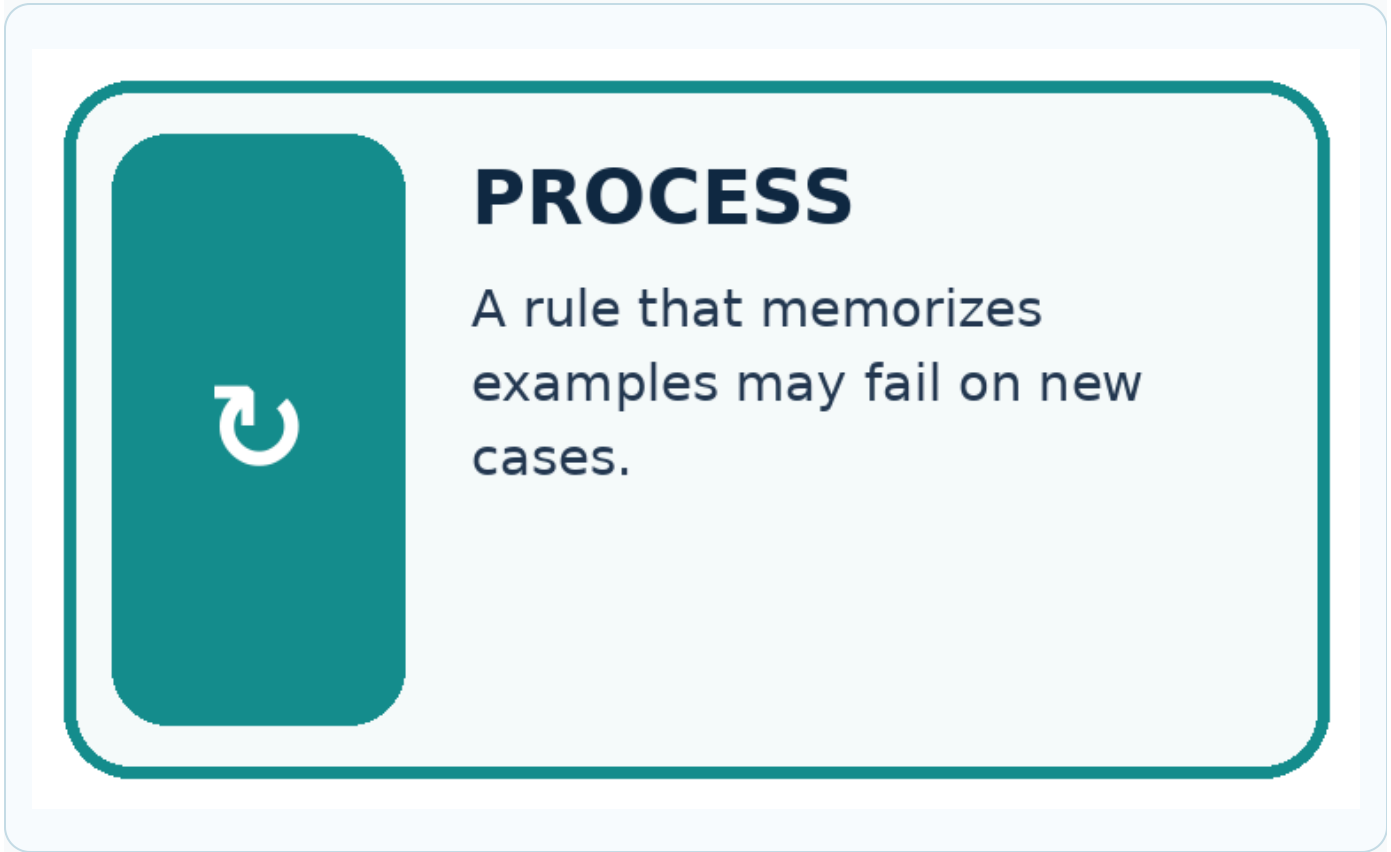
VISUAL **MECHANISM**

Input → **Process** → **Output**

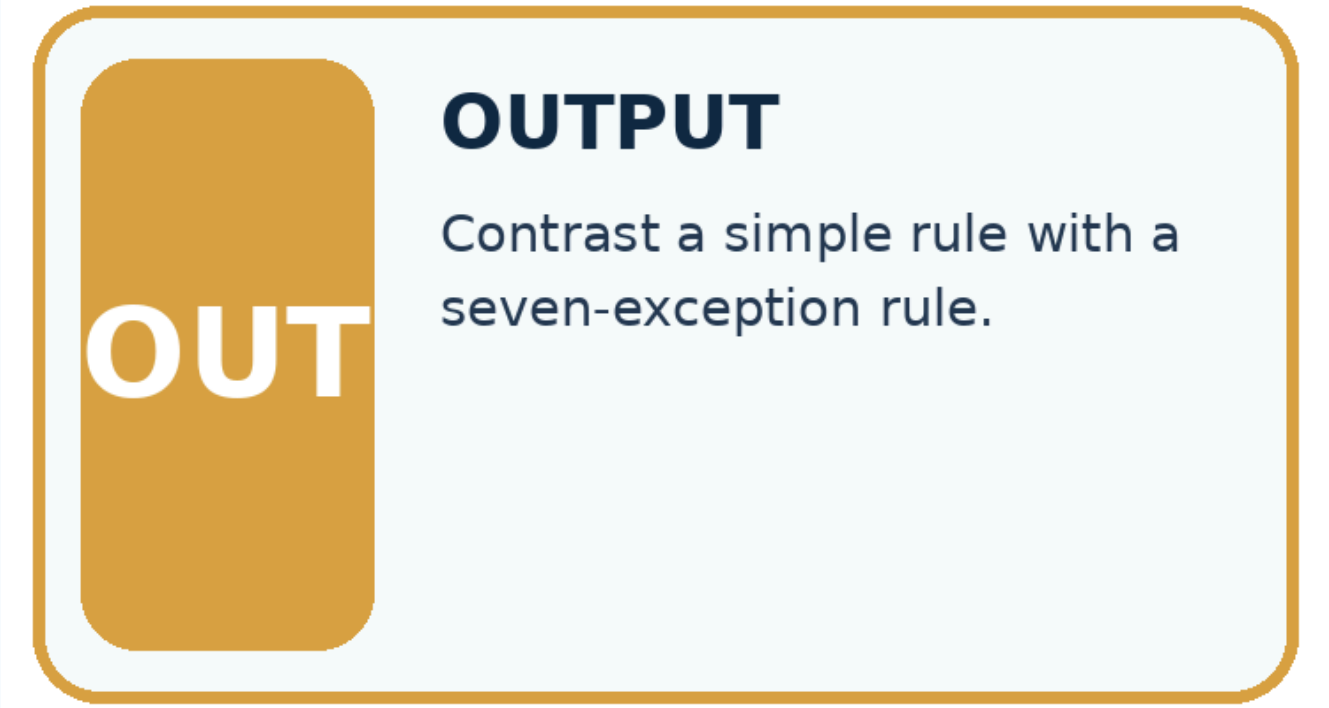
Trace the information. At each stage, ask what changed and how you could test it.


IN
INPUT
Contrast a simple rule with a seven-exception rule.

↓


↻
PROCESS
A rule that memorizes examples may fail on new cases.

↓



Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

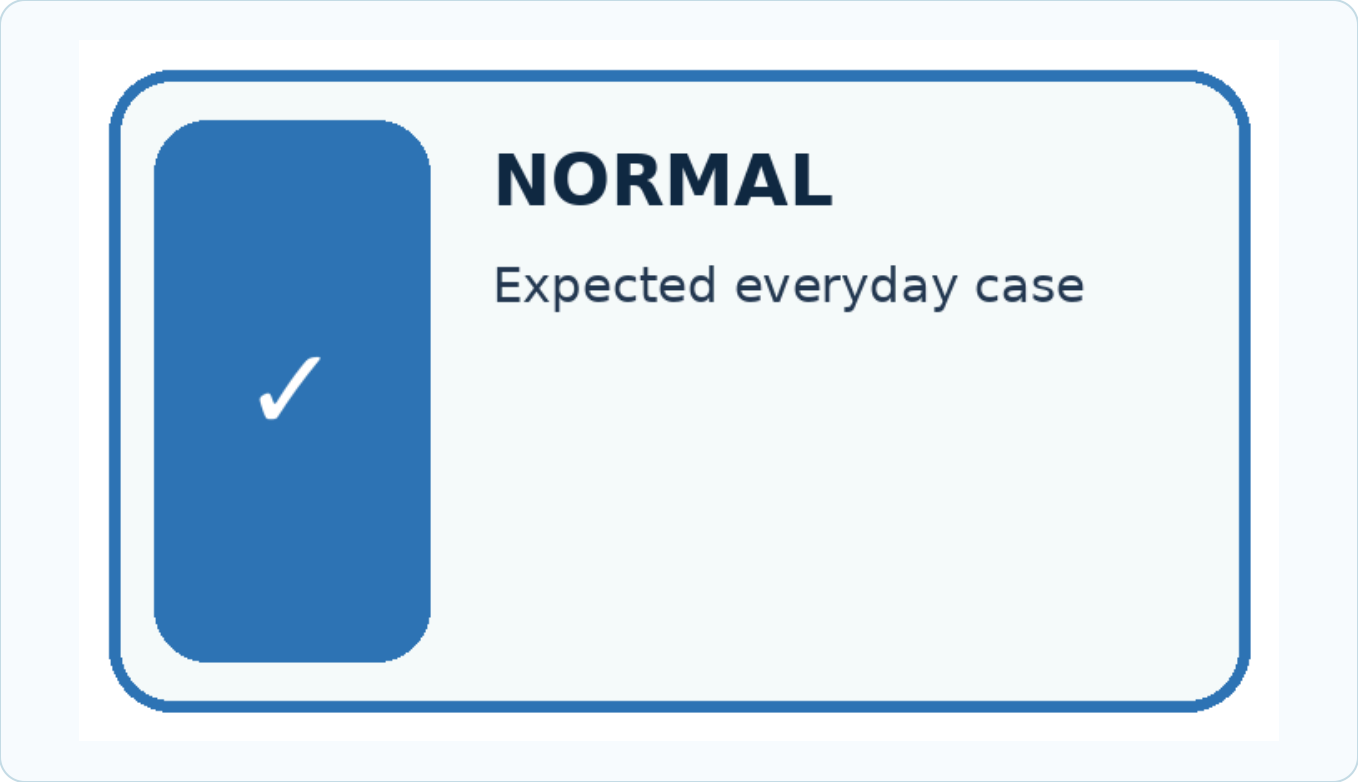
A closer explanation

Connect the picture and the explanation

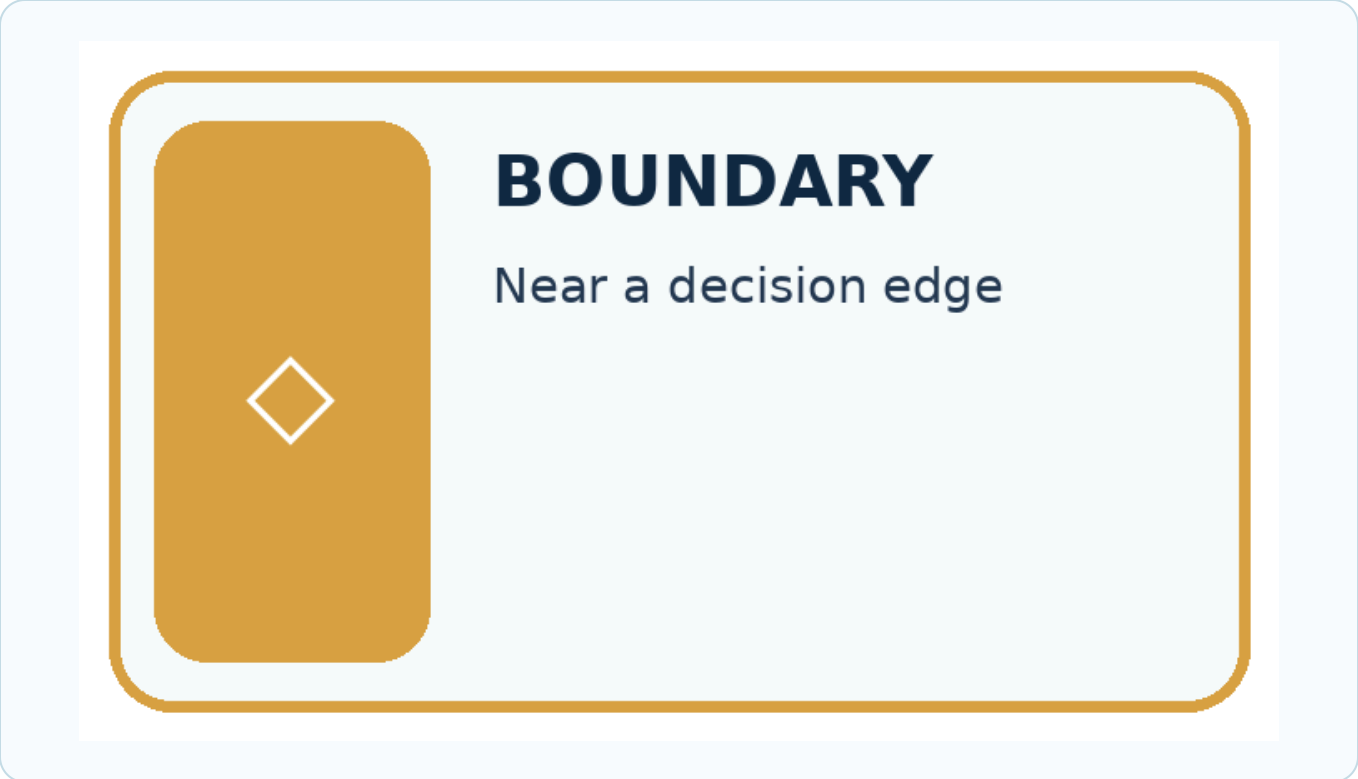
Test the claim against a changed comparison

> A more flexible candidate can win when its valid evidence is better.

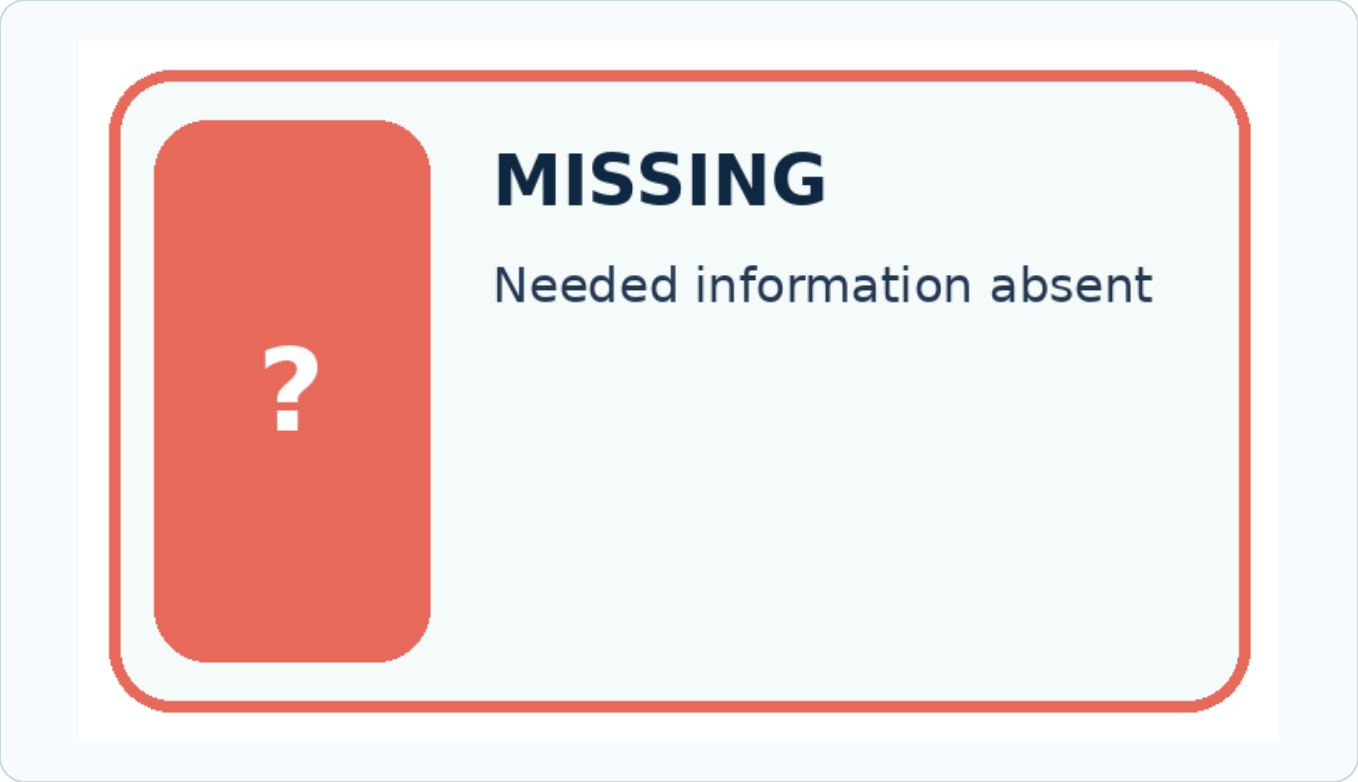
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

> **Write or draw your reasoning and evidence.**

Predict the supplied changed case

- Change: Complex's validation error becomes 2 on a newly documented, fair development comparison.

▼ Check after predicting

Prediction: The stated rule now selects Complex.

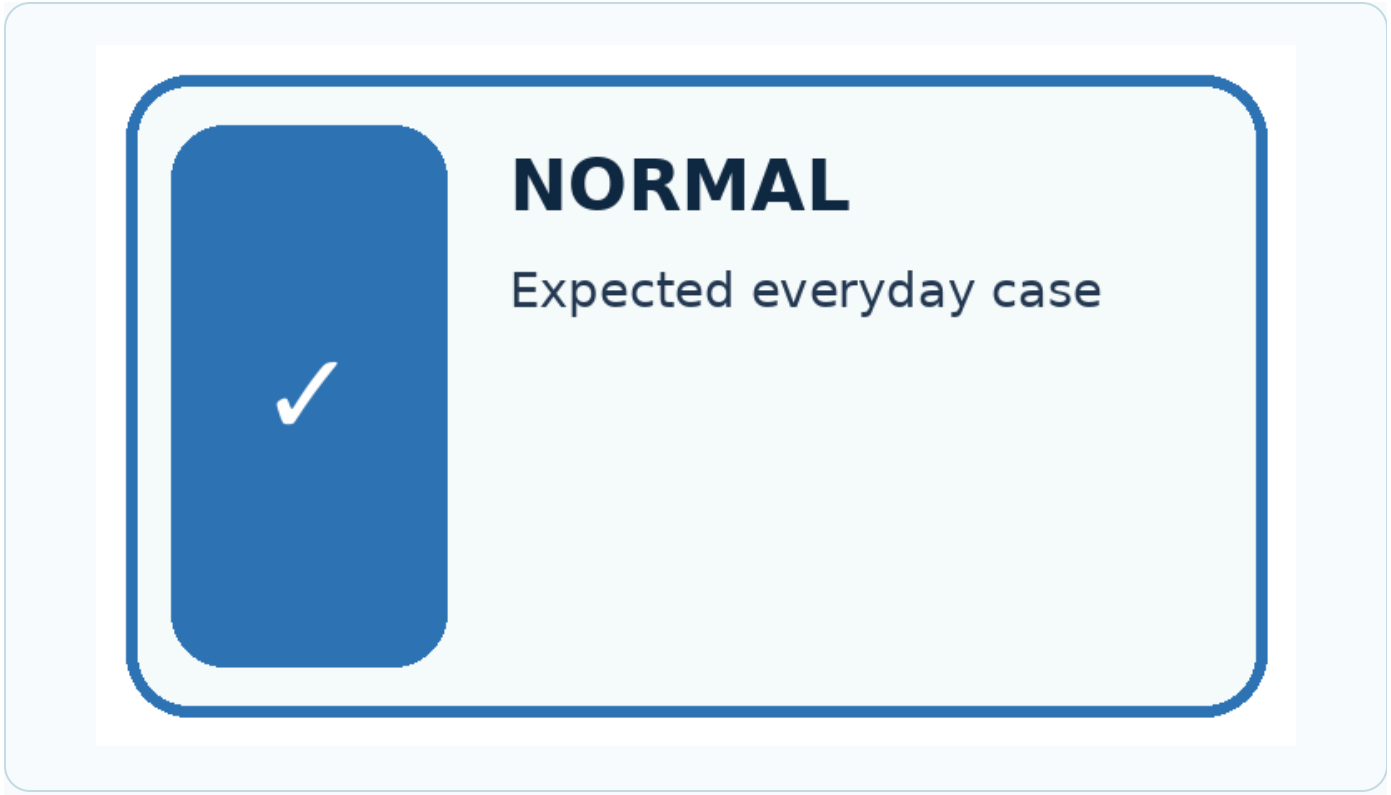
Reason: The rule compares validation errors; it does not ban complex models.

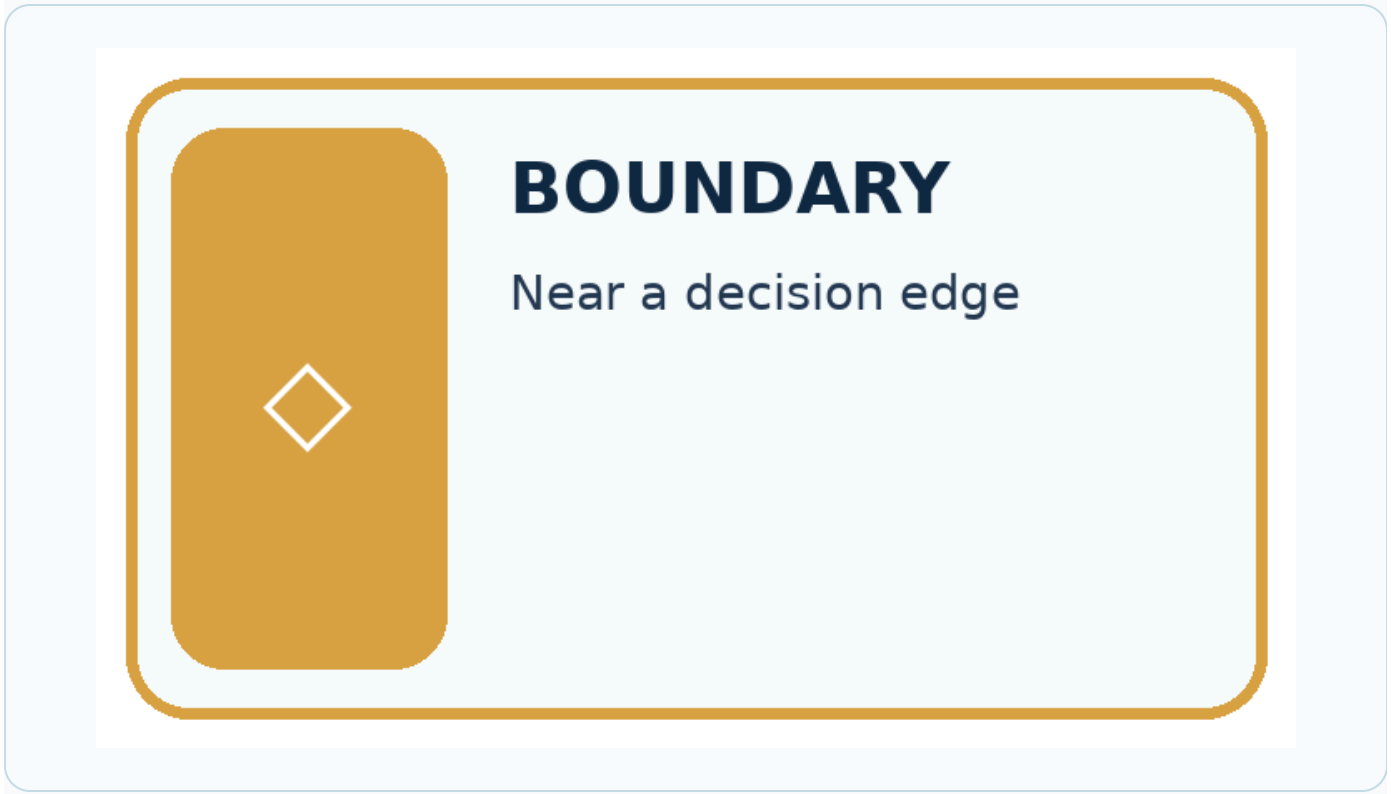
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

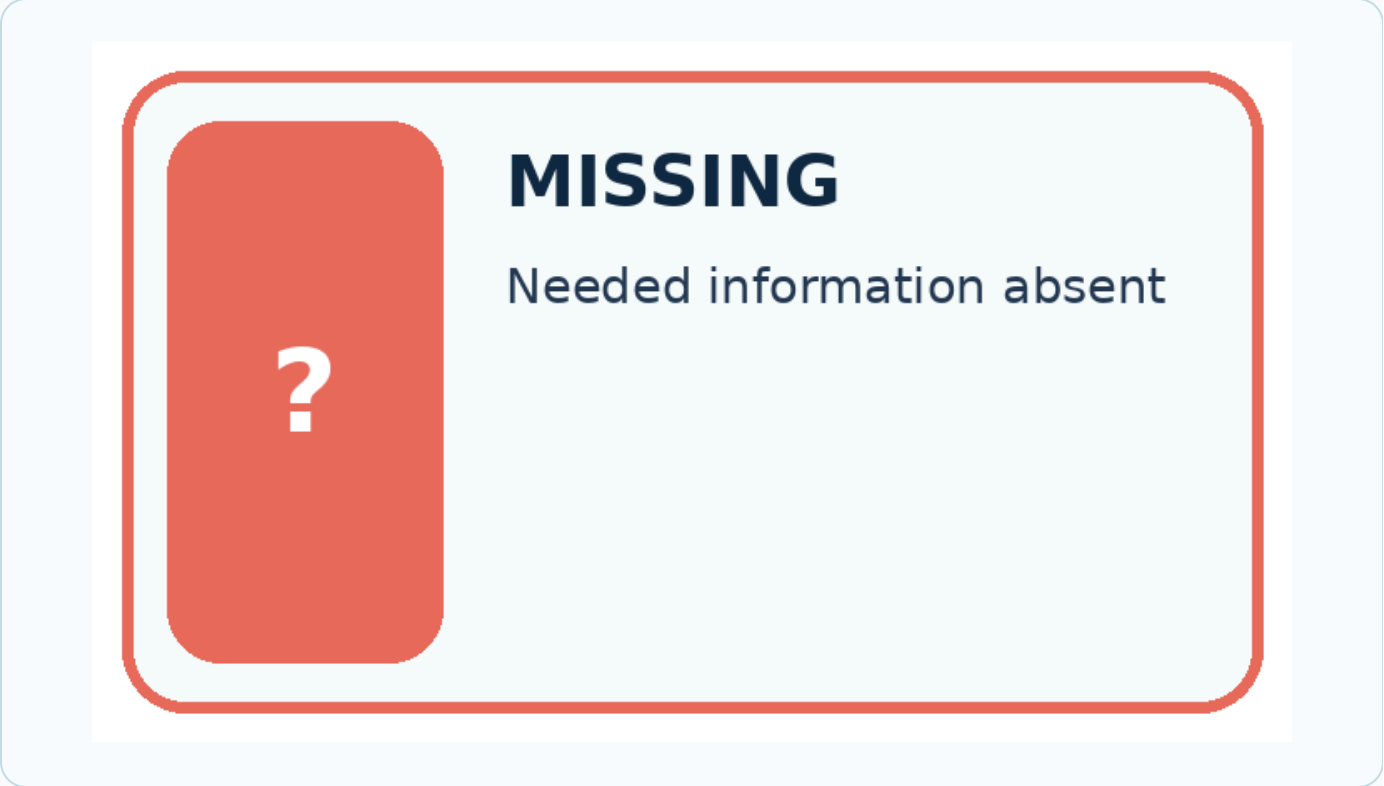
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Interpret the training trap, then a separate pictured comparison

> Use the validation column without claiming simplicity always wins.

Read the opening comparison

Overfitting and model capacity at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Constant	Training 5	Validation 6
Line	Training 2	Validation 3
Flexible curve	Training 0	Validation 8

The curve result is consistent with overfitting, but a gap alone does not prove its cause. A changed population, bad labels or a split error could also produce a gap. Inspect the procedure before claiming a diagnosis. The constant may miss useful structure, but these numbers alone do not identify the true relationship.

Question a gap-only diagnosis

Does a training-validation gap alone prove overfitting? Explain the steps or evidence behind your response.

Keep the two supplied records separate

Keep the two records distinct. The opening Constant/Line/Flexible comparison has validation errors6,3,8. The separate pictured Simple/Medium/Complex comparison below has6,3,7. Both select their middle candidate under the stated rule, but the last error is different. None of these numbers is an untouched final-test result.

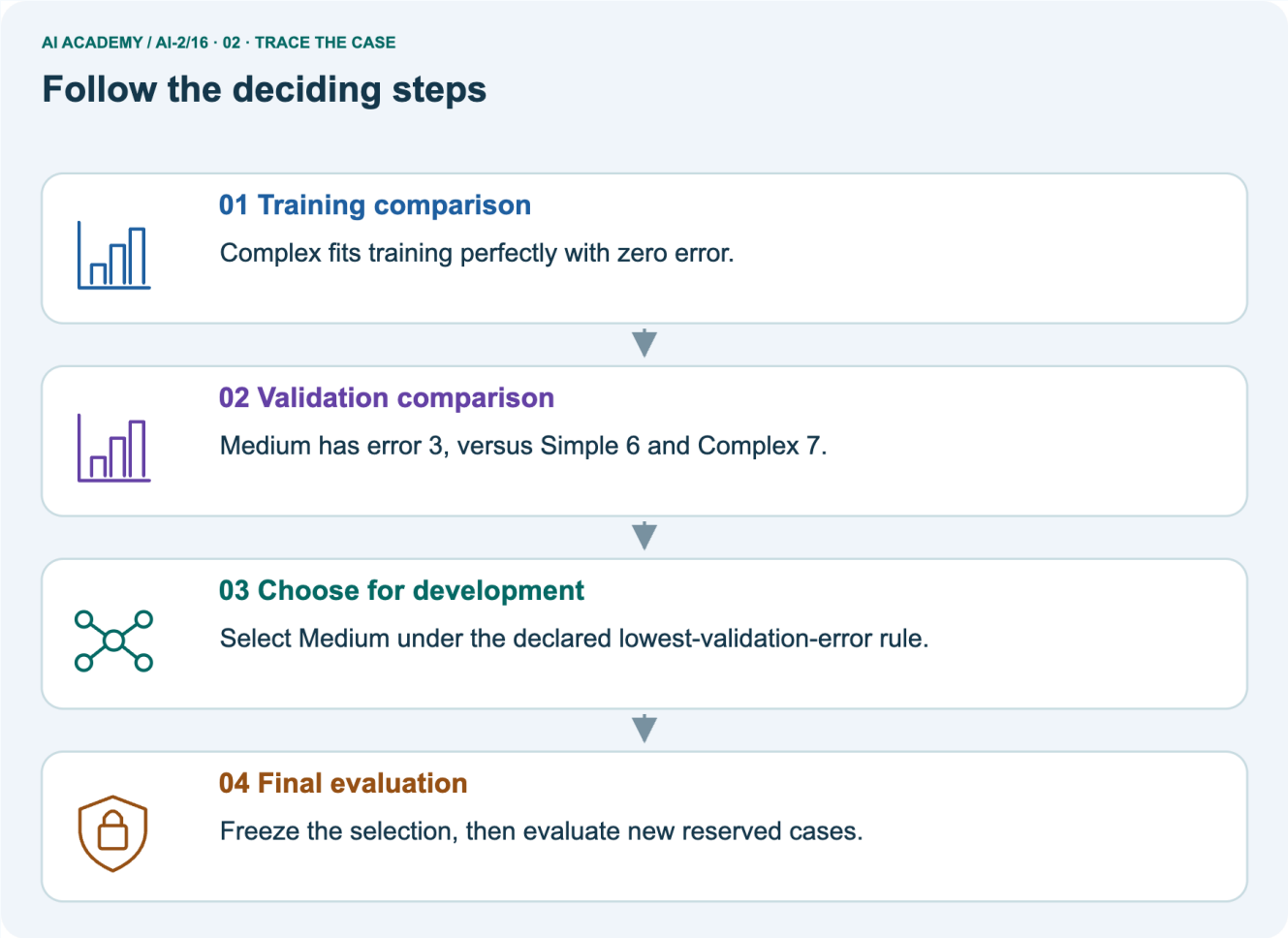
Introduce the separate pictured candidates

A fictional project compares three book-length **prediction** models on the same development sets.

Mean absolute errors in pages: Simple **training** 5, validation 6; Medium **training** 2, validation 3; Complex **training** 0, validation 7.

Select lowest validation error; untouched final tests are not used for selection.

Trace development selection and final-test protection



Follow the labelled picture with the matching explanation. Open the vector diagram

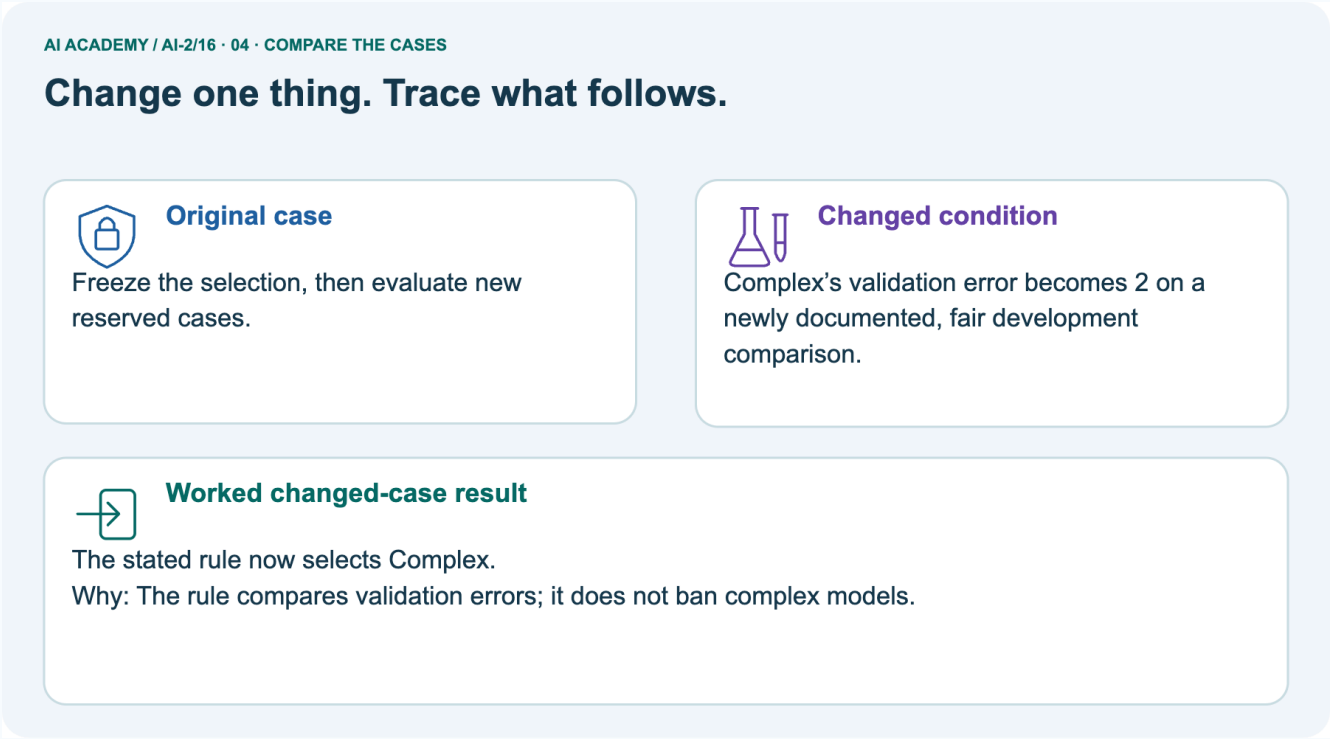
A flexible model can fit quirks of its **training** set without predicting new examples well. Here Complex's **training** success occurs alongside the worst validation error.

The numbers are invented comparisons, not a claim that **complexity** always hurts. Choosing Medium uses development **evidence**; its validation score must not be advertised as an untouched final-test result.

Predict a newly documented development result

- Change one thing: Complex's validation error becomes 2 on a newly documented, fair development comparison. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the original choice and remaining uncertainty

- Read the labelled picture: Does zero **training** error justify choosing Complex in the original record?
- A single small validation set may be noisy; selection **evidence** does not prove future population performance.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

WEEK 16 • WORKED EVIDENCE

1. Simple rule: difficulty ≤ reader level + 1.
2. Memorized rule: seven title-specific exceptions.
3. Both fit the seen books; only new books reveal generalization.
4. Prefer the simplest rule that performs well on fair validation cases.

1. Simple rule: difficulty ≤ reader level + 1.

2. Memorized rule: seven title-specific **exception**s.

3. Both fit the seen books; only new books reveal **generalization**.

4. Prefer the simplest rule that performs well on fair validation cases.

Recalculate, retrace, or restate the reasoning in your own representation:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Overfitting Intuition

The facts we will use

AI ACADEMY / AI-2/16

Training and validation errors

	Training MAE pages	Validation MAE pages
Simple	5	6
Medium	2	3
Complex	0	7

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-2/16

Overfitting Intuition

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Training comparison

Complex fits training perfectly with zero error.



2. Validation comparison

Medium has error 3, versus Simple 6 and Complex 7.



3. Choose for development

Select Medium under the declared lowest-validation-error rule.



4. Final evaluation

Freeze the selection, then evaluate new reserved cases.

CHANGE ONE THING

Complex's validation error becomes 2 on a newly documented, fair development comparison.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Training comparison	Complex fits training perfectly with zero error.
2. Validation comparison	Medium has error 3, versus Simple 6 and Complex 7.
3. Choose for development	Select Medium under the declared lowest-validation-error rule.
4. Final evaluation	Freeze the selection, then evaluate new reserved cases.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

A gap suggests questions to investigate, not a complete diagnosis

> Check the split and data before claiming a model memorised noise.

Complexity is one experimental factor

A simpler model is not always better. If a flexible candidate has better valid **evidence** under the declared rule, keep that **evidence** visible.

Model **capacity** is one part of an experiment, alongside features, sample size, label quality and the split.

Name further evidence to inspect

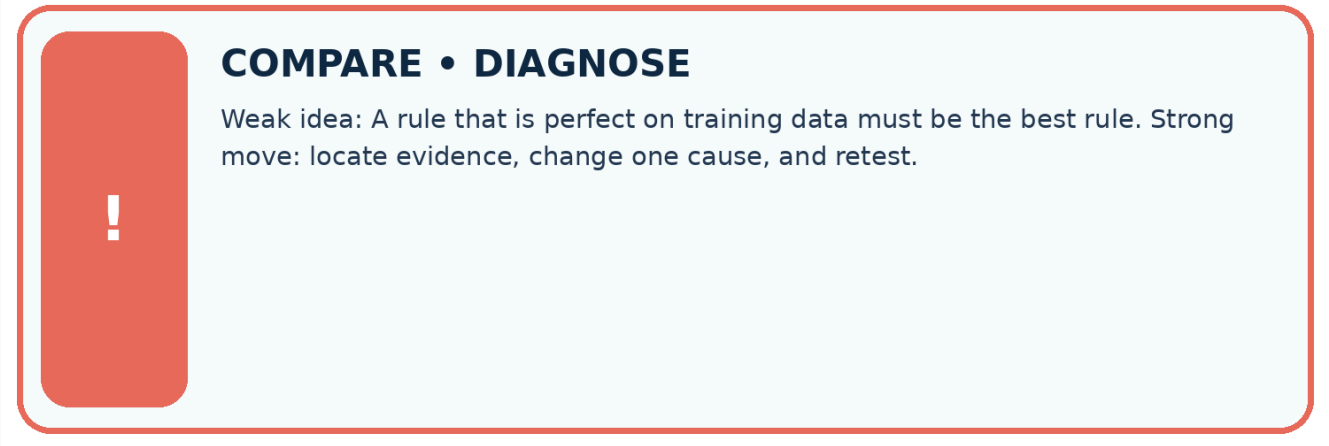
What extra evidence would you inspect before concluding a model memorized noise?

▼ **Earlier wording, alternatives and picture transcripts**

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.



Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

> **Write or draw your reasoning and evidence.**

A closer explanation

MAIN LESSON · SECTION 8

Plan one controlled development comparison

> Keep training-only fitting, fixed validation cases and the metric visible.

Change one declared factor and preserve valid cases

Change one development factor at a time: simplify the candidate, improve data coverage or adjust a declared regularization setting. Refit on **training** and compare on validation.

Do not remove difficult valid examples or repeatedly search the test for a favorable result.

Interpret a gap in W2

Use workbook W2 for the connected example "**Overfitting** and model **capacity**". First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Repair an unsupported final-answer claim

- Use the worked case for **Overfitting** and model **capacity**. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured development choice

Does zero **training** error justify choosing Complex in the original record?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.

TRACE THE LOGIC

Predict the result. Trace every condition or calculation. Compare expected with actual. Explain the first difference.

SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Practise with less help: Interpret a gap

Check your reasoning

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Audit validation MAE from complete prediction records

> Retain every per-book error and distinguish supplied summaries from calculated evidence.

The full four-book validation record

Validation ID · Reference pages · Constant prediction · Line prediction · Flexible prediction

> **Validation ID:** V1
> **Reference pages:** 10
> **Constant prediction:** 22
> **Line prediction:** 13
> **Flexible prediction:** 18

> **Validation ID:** V2
> **Reference pages:** 14
> **Constant prediction:** 22
> **Line prediction:** 17
> **Flexible prediction:** 22

> **Validation ID:** V3
> **Reference pages:** 18
> **Constant prediction:** 22
> **Line prediction:** 21
> **Flexible prediction:** 26

> **Validation ID:** V4
> **Reference pages:** 22

> **Constant prediction:** 22
>
> **Line prediction:** 25
>
> **Flexible prediction:** 30

Predict and calculate under a fixed comparison

1. These four invented validation records supply the opening comparison’s predictions. Keep the candidate names and all four rows fixed. Before calculating, predict which has the smallest average error. For each candidate write four absolute errors, their sum and sum/4 with units pages. Check the resulting validation MAEs against6,3,8. Training MAEs5,2,0 are supplied summary evidence; no training predictions are provided, so you cannot reconstruct those training scores here. This is an arithmetic audit, not a fitting experiment or final test. Keep an unresolved discrepancy visible rather than editing a reference to improve a score.

Choose and propose a controlled next step

Using the audited results, select under smallest validation MAE and write one fair next development comparison. Change one candidate or training procedure at a time, fit using training only and compare on the same validation cases and metric. Log any repeated validation-driven choices and reserve an untouched final set. Do not claim a good validation score proves future performance or that the four rows establish the cause of the flexible candidate’s poor result. Reserve A/B/C and its simplicity tie rule for independent W4.

Keep the record and explanation

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

Test / input: boundary

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: missing

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

Test / input: fresh transfer

Prediction: _____

Actual: _____

Evidence / calculation: _____

Change or keep?: _____

What did the hardest test teach you that the normal test did not?

Write or draw your reasoning and evidence.

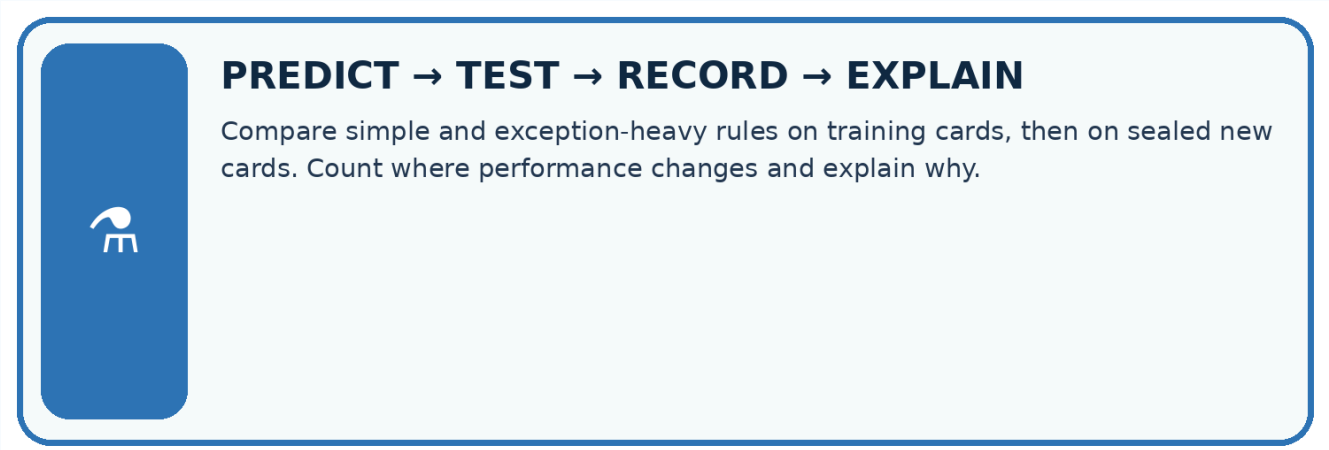
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.



Audit the supplied four-book validation predictions: retain each absolute error, sum and denominator, compare with the supplied **training** summaries and explain a controlled next step. No sealed-card dataset or real model fitting is supplied for this investigation.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: apply the validation rule and simplicity tie-break

> Do not select using zero training error.

The new A/B/C comparison

Models A **training** 4 validation 5, B **training** 1 validation 2, C **training** 0 validation 2. Lowest validation MAE then simpler; B is simpler than C. Choose and justify.

Units, rule and evidence scope

For this independent comparison all numbers are MAEs in pages on the same respective training and validation sets. Lower validation MAE wins; use the stated simplicity tie-break only when validation errors are equal. No final-test results are supplied.

Your first attempt and help record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Apply independently

Week 16 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

Rule A memorises six **training** book IDs. Rule B uses a relevant topic feature. Why might A fail on a new ID? What test is needed?

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Avoid title-by-title patches; improve features or thresholds that transfer.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Rule A memorises eight **training** IDs and gets 8/8 **training**, 2/8 validation. Rule B uses genre and gets 6/8 **training**, 6/8 validation. Which is preferable on this **evidence**, and what test remains?

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Compare recommender revisions on a declared task

> Keep numerical prediction, category decisions and preference ranking distinct.

This week’s concrete portfolio contribution

Record the actual portfolio task and metric before comparing revisions. A numeric book-length prediction can use MAE in pages; an eligibility classifier needs a declared classification measure; an overall preference ranking is not automatically either one. Preserve related-record splits, training-only fitting and a change log. A title-specific fix is not evidence of transfer. Use fictional profiles and keep final cases separate from development choices.

Compare recommender revisions on fixed development cases, reserving new cases for the final presentation.

Keep the checkpoint and human choice visible

THIS WEEK'S CHECKPOINT Avoid title-by-title patches; improve features or thresholds that transfer.

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

> **Write or draw your reasoning and evidence.**

What should remain under **human control**?

> **Write or draw your reasoning and evidence.**

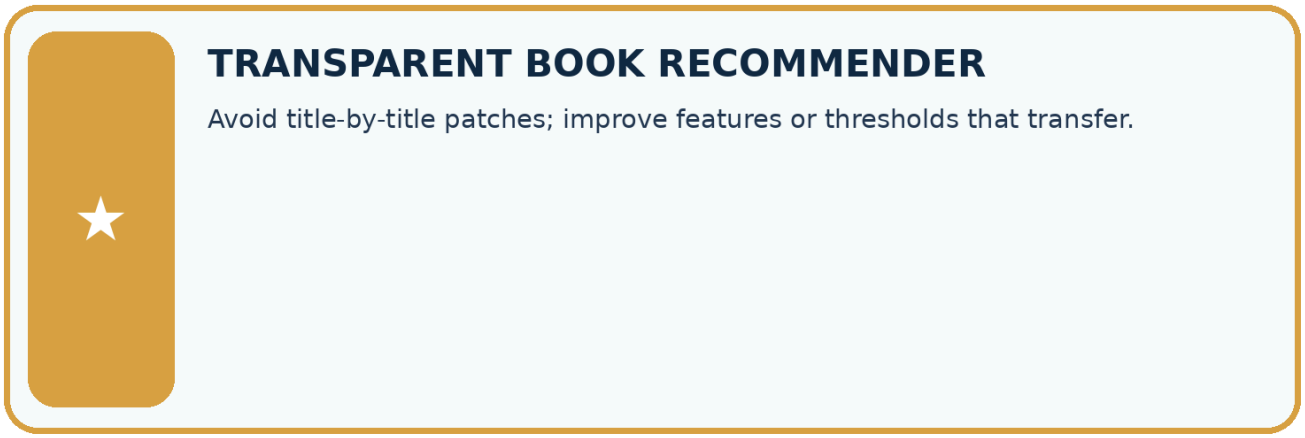
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits. Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



Evidence for your project

MAIN LESSON · SECTION 16

Finish: justify the comparison without overstating the cause

> Name the metric, development evidence and what remains untested.

Your explanation without notes

- Close your notes. Explain this goal in your own words: Compare **training** and validation **evidence** for model **capacity**

My deciding step, evidence and one remaining question

Choose a changed-case return

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	My own example	Not this / boundary
overfitting	Adapting too closely to development examples so performance does not transfer well to new relevant cases.	
generalize	Perform usefully on new relevant cases beyond those used to develop a system.	
complexity	The amount of structure, work or resources a method requires, measured in a stated way.	
exception	A condition that requires handling outside the usual path.	

Terms for the connected explanation

Word	Meaning
Capacity	Flexibility of a model to represent patterns

Word	Meaning
Underfitting	Failing to capture useful structure
Generalization	Performance on relevant unseen examples

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY LAB

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.



2

GENERALIZE

work on new relevant cases

3

COMPLEXITY

number of conditions or exceptions



Words used in the closer explanation

▼ **Choose a stronger thinking challenge**

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.
- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.

- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.

- My strongest **evidence** of independent understanding:

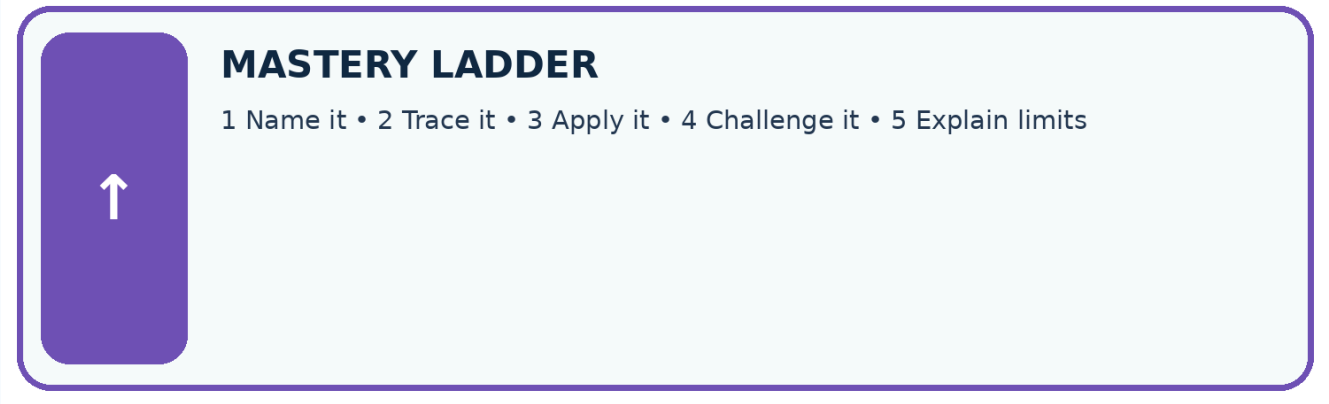
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference: check your explanation

Reference: check your explanation

> Use after practice. Apply independently is the selected independent case.

Check the evidence and its limit

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check and record

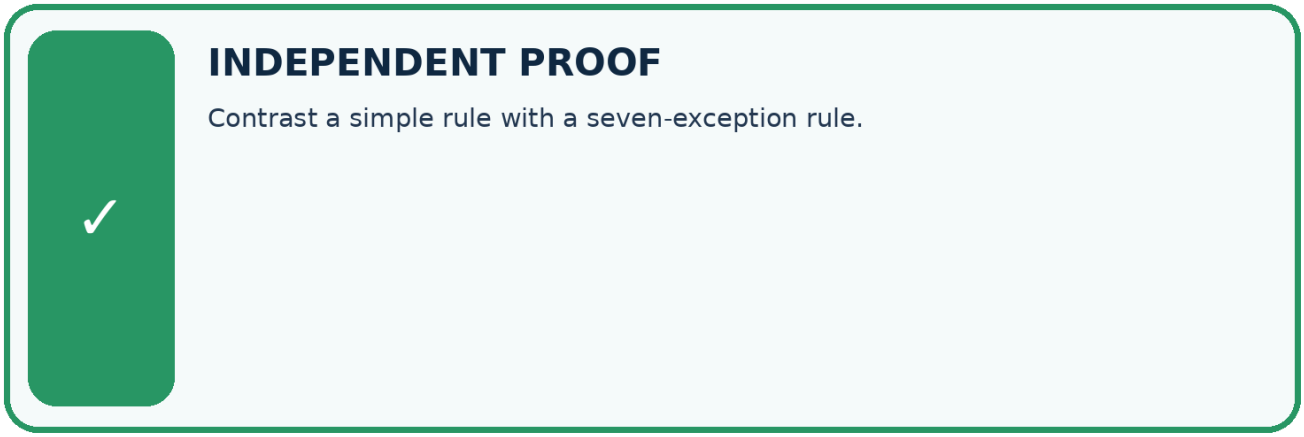
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for explaining this week's idea

Use after practice. Apply independently is the selected independent case.



EXPLANATION CHECK Keep the metric, units, split and declared selection rule visible; development evidence does not prove a causal diagnosis or future performance.

Write or draw your reasoning and evidence.

▼ **Extra practice: Perfect recall can hide weak transfer**

REFERENCE · SECTION 13

Extra practice: Perfect recall can hide weak transfer

> Try the question before opening its worked explanation.

Read the supplied case and question

Rule A memorises six **training** book names. It scores 6/6 familiar and 2/6 new. Rule B uses relevant traits but is untested.

What does the **evidence** support?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

A shows weak transfer. B might generalise better but requires testing; simplicity is not proof.

Evidence or stage	Value or result
Familiar books	6/6
New books	2/6

▼ Earlier wording, alternatives and picture transcripts

Week 16 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Perfect recall can hide weak transfer

Rule A memorises six **training** book names. It scores 6/6 familiar and 2/6 new. Rule B uses relevant traits but is untested.

What does the **evidence** support?

A shows weak transfer. B might generalise better but requires testing; simplicity is not proof.

▼ **Reference: original lesson checklist**

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Compare training and validation evidence for model capacity	Explain the deciding step, show a trace or test, and state one limit.
Explain overfitting without assuming every gap proves it	Explain the deciding step, show a trace or test, and state one limit.
Use a declared selection rule instead of perfect training fit	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional project compares three book-length **prediction** models on the same development sets.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional project compares three book-length **prediction** models on the same development sets.
- Which labelled step matches this detail? Freeze the selection, then evaluate new reserved cases.
- 1. Final evaluation
- 2. **Training** comparison
- 3. Choose for development

- 4. Validation comparison

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Final evaluation Freeze the selection, then evaluate new reserved cases. A flexible model can fit quirks of its **training** set without predicting new examples well.

Here Complex's **training** success occurs alongside the worst validation error. The numbers are invented comparisons, not a claim that **complexity** always hurts.

Choosing Medium uses development **evidence**; its validation score must not be advertised as an untouched final-test result.

Changed-case question

- Changed-case practice: Complex's validation error becomes 2 on a newly documented, fair development comparison. Which conclusion is supported by the supplied facts?

- 1. We can decide the result without examining the changed **input** or the procedure.

- 2. This one example guarantees correct results for every future **input**.

- 3. The stated rule now selects Complex.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: The stated rule now selects Complex. The rule compares validation errors; it does not ban complex models.

Repair and save your work

Review Choose Informative Questions (ai-2/15). Goal: Fit a root question and leaf **prediction**s from **training** examples. Short practice: Choose the best original question and explain why its more balanced split matters here..

Return to this week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

Review Human Classifier Simulation (ai-2/10). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Why are J and M initially treated differently even though both are illustrated mysteries?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

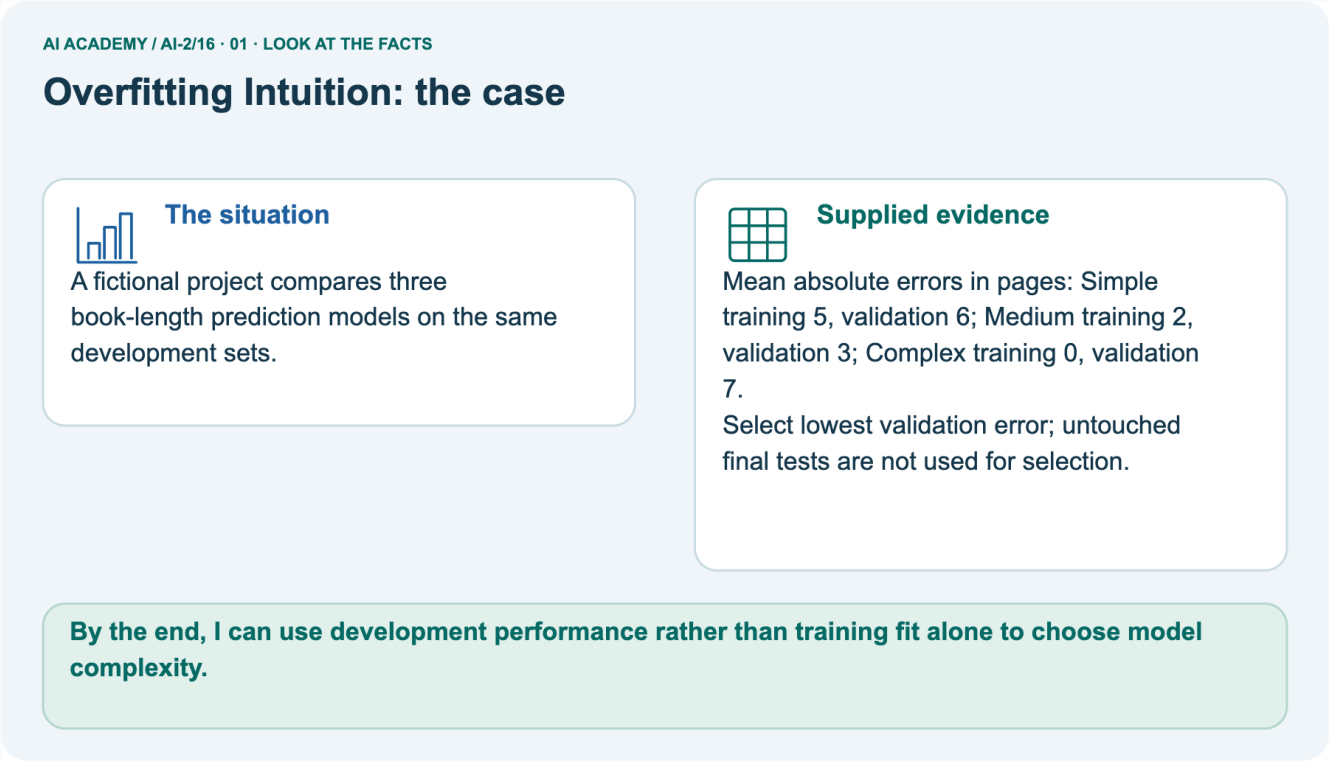
▼ Reference: the six-picture journey

REFERENCE · SECTION 18

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Overfitting Intuition: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

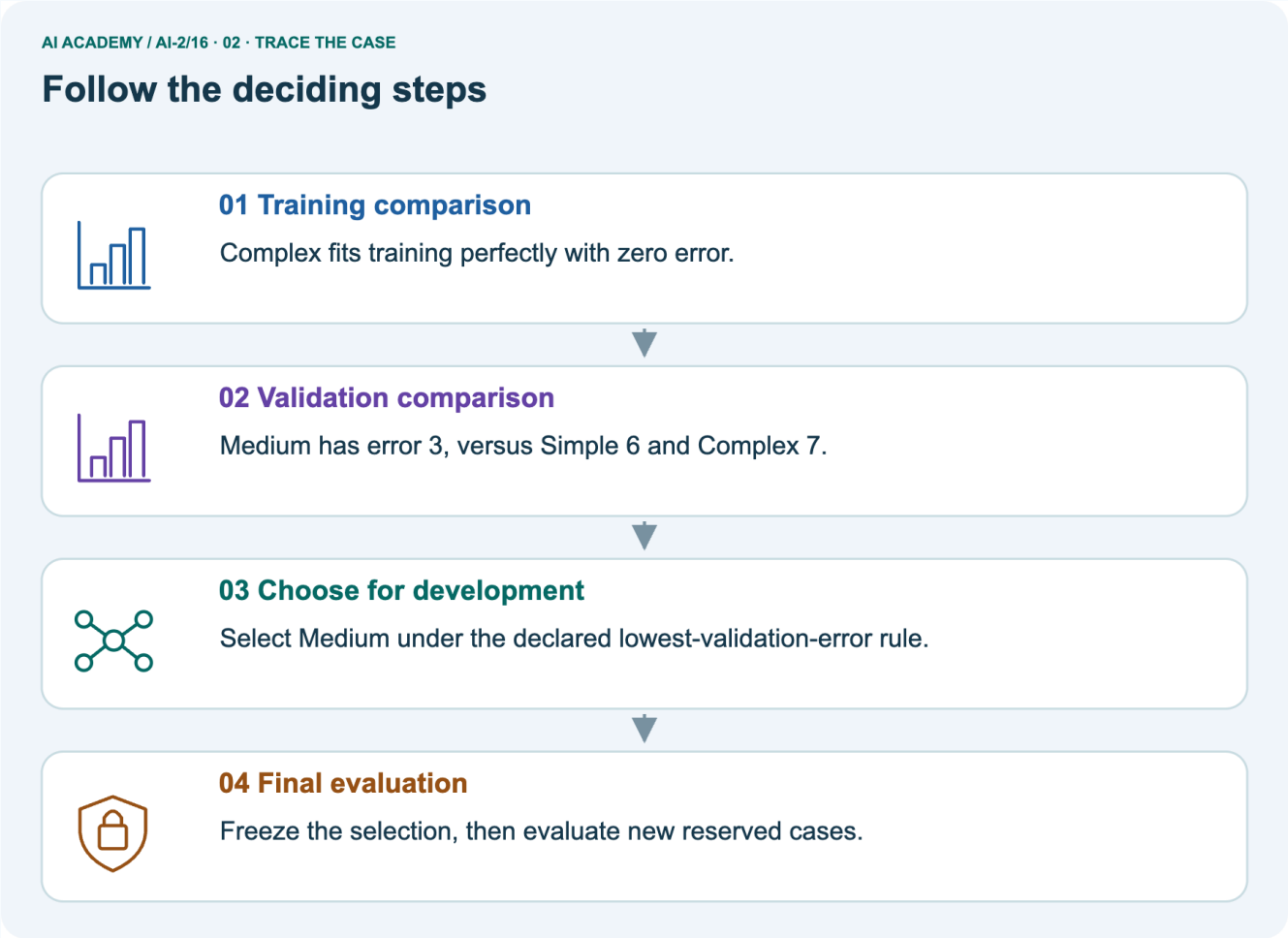
A fictional project compares three book-length **prediction** models on the same development sets.

Supplied facts: Mean absolute errors in pages: Simple **training** 5, validation 6; Medium **training** 2, validation 3; Complex **training** 0, validation 7.

Select lowest validation error; untouched final tests are not used for selection.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

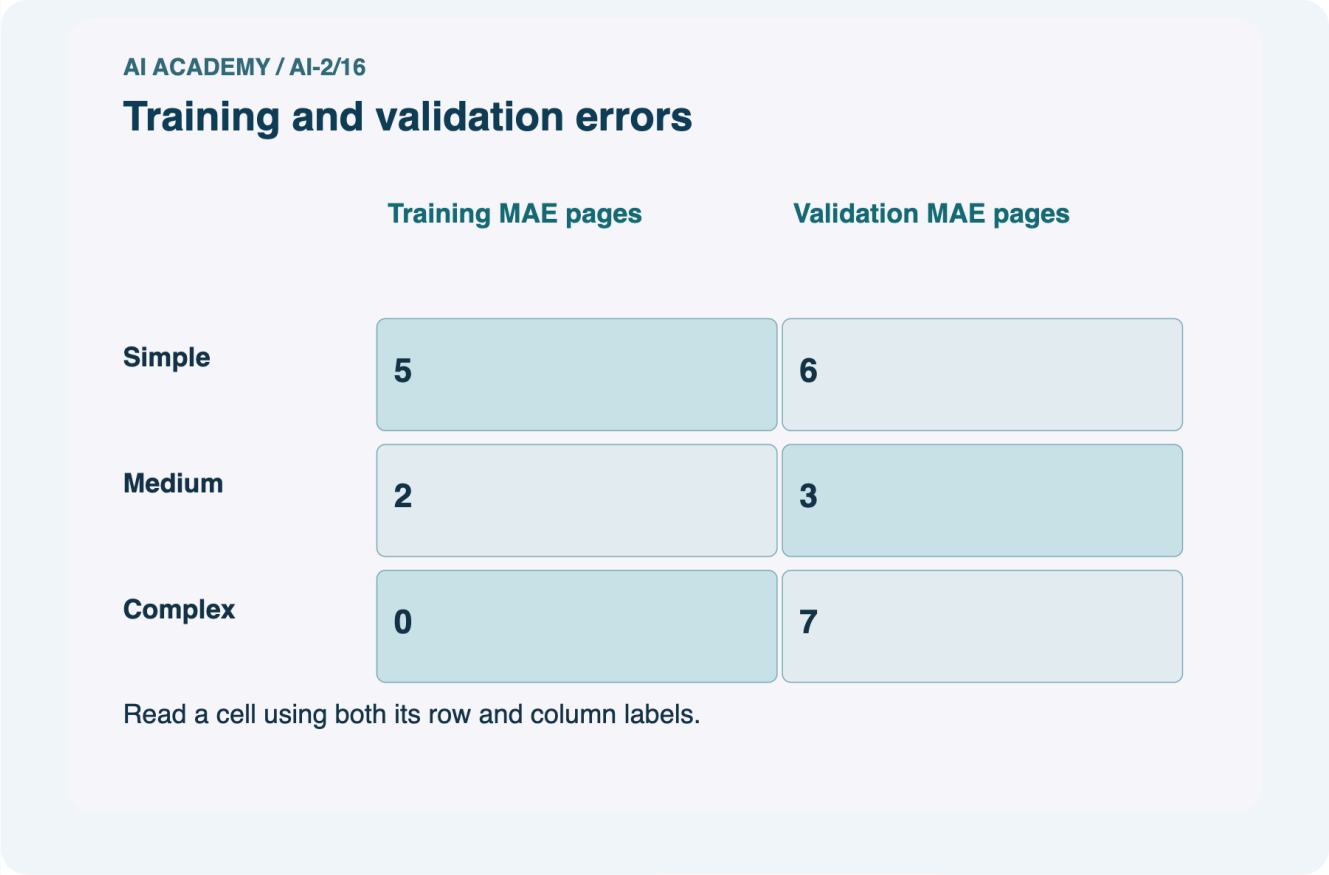


Follow the labelled picture with the matching explanation. Open the vector diagram

Training comparison: Complex fits **training** perfectly with zero error. → Validation comparison: Medium has error 3, versus Simple 6 and Complex 7. → Choose for development: Select Medium under the declared lowest-validation-error rule. → Final evaluation: Freeze the selection, then evaluate new reserved cases.

Point to each numbered stage and explain which recorded **input** it uses.

03. Training and validation errors

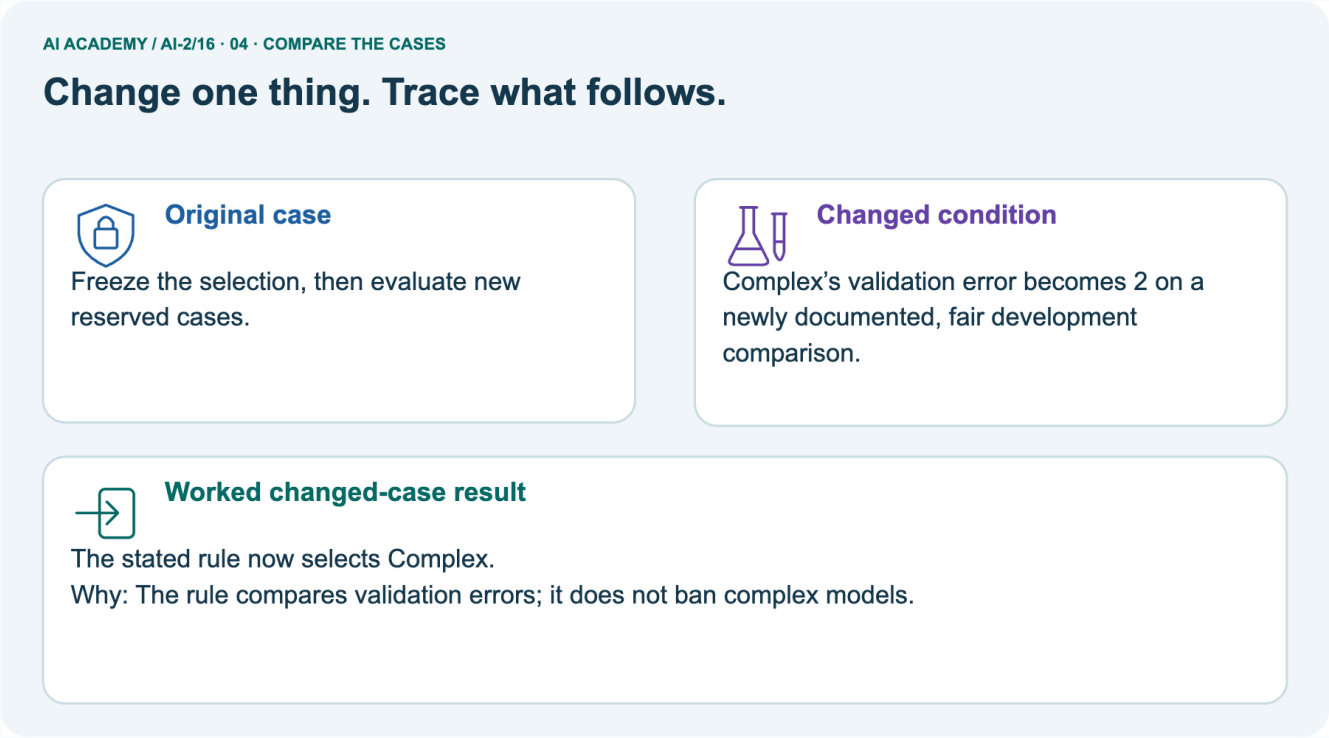


Follow the labelled picture with the matching explanation. Open the vector diagram

Training and validation errors. Simple: **Training** MAE pages = 5, Validation MAE pages = 6. Medium: **Training** MAE pages = 2, Validation MAE pages = 3. Complex: **Training** MAE pages = 0, Validation MAE pages = 7.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



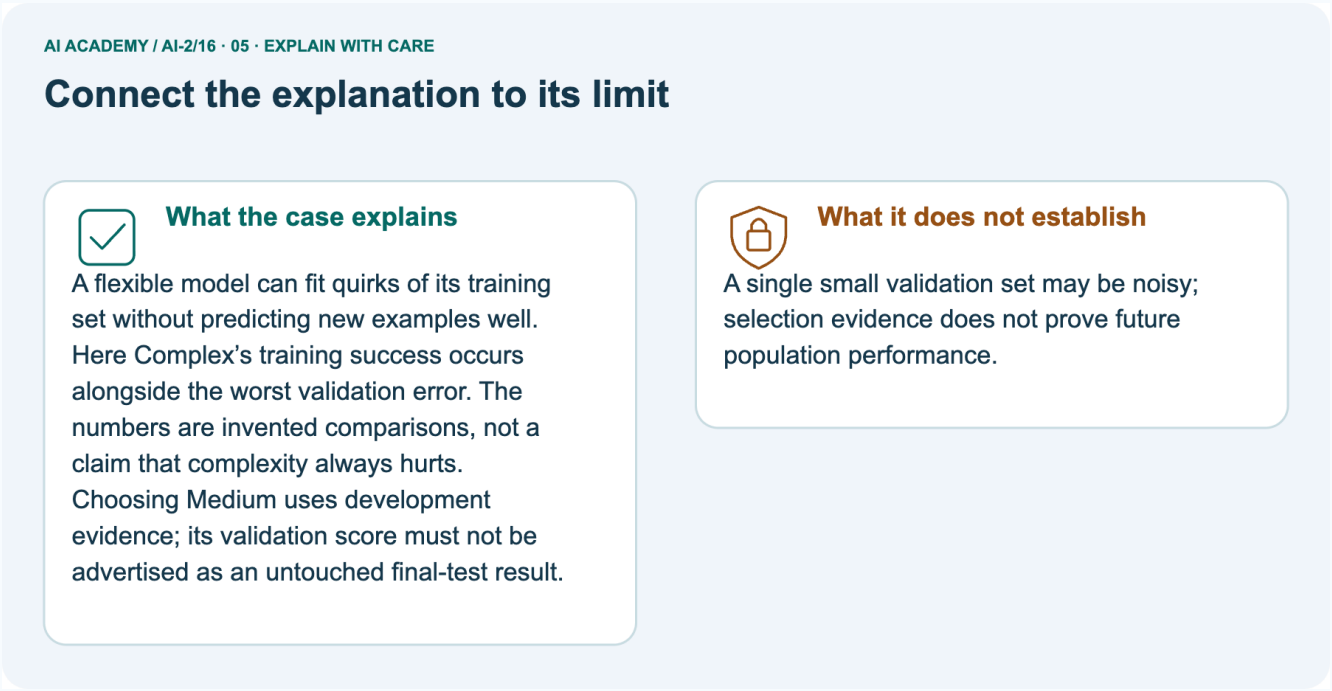
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Freeze the selection, then evaluate new reserved cases. Changed condition: Complex's validation error becomes 2 on a newly documented, fair development comparison.

Expected result: The stated rule now selects Complex. Reason: The rule compares validation errors; it does not ban complex models.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

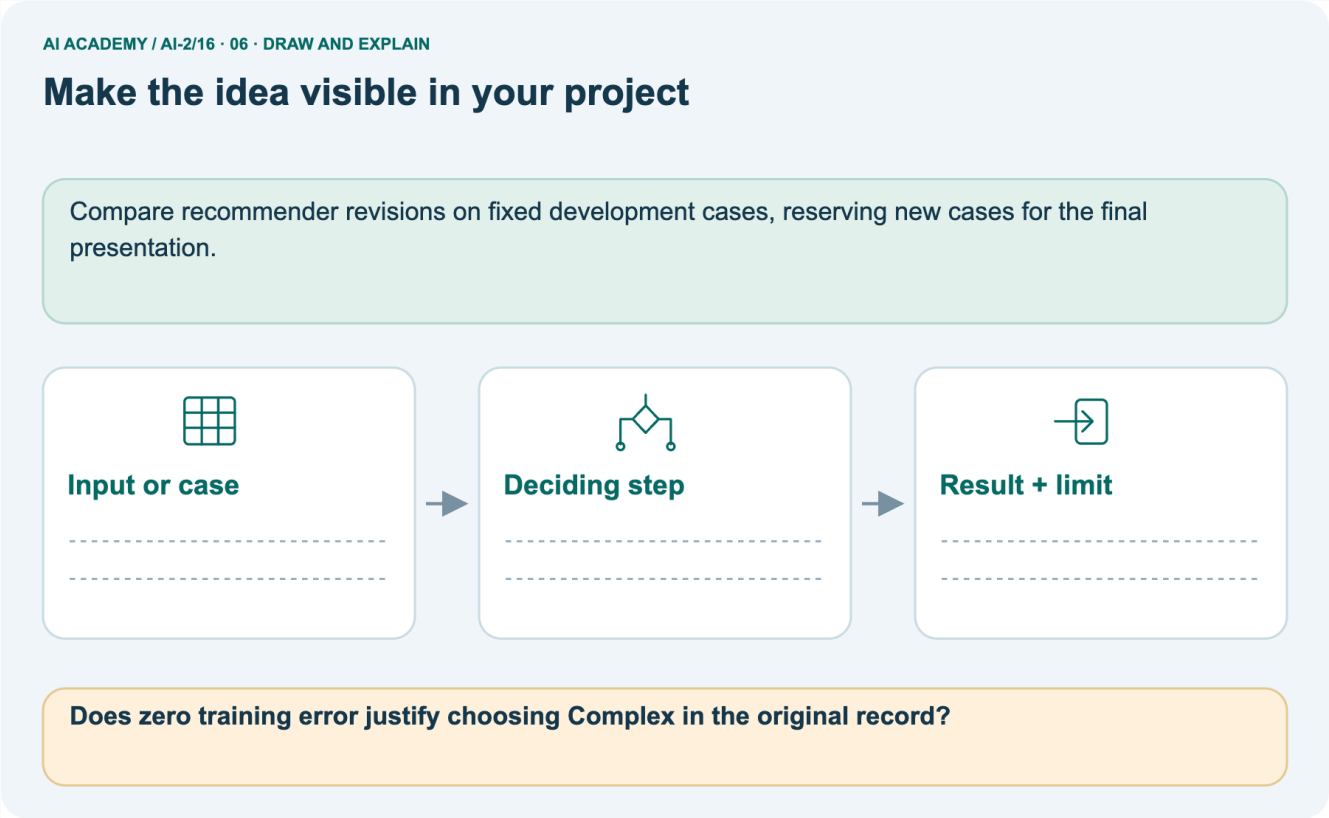
A flexible model can fit quirks of its **training** set without predicting new examples well. Here Complex's **training** success occurs alongside the worst validation error.

The numbers are invented comparisons, not a claim that **complexity** always hurts. Choosing Medium uses development **evidence**; its validation score must not be advertised as an untouched final-test result.

Limit: A single small validation set may be noisy; selection **evidence** does not prove future population performance.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Compare recommender revisions on fixed development cases, reserving new cases for the final presentation. Draw your **input**, deciding step, result and limit.

Supported question: Does zero **training** error justify choosing Complex in the original record?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Overfitting Intuition: the case

02. Follow the deciding steps

03. Training and validation errors

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.